ALGEBRA 2 · EXPONENTIAL & LOGARITHMIC FUNCTIONS

Exponential & Log Mirror

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Exponential & Log Mirror** on Mr. K's Classroom: Algebra 2 → Exponential & Logarithmic Functions → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. If $2^3 = 8$, what is $\log_2 8$?

   3

2 Explore use the tool and record what you see

Matching points

Move the input slider and record both points.

Exponential form	Log form	Point on $y = b^x$	Point on $y = \log_b x$
$2^3 = 8$	**$\log_2 8 = 3$**	**(3, 8)**	**(8, 3)**
$2^{-1} = \frac{1}{2}$	**$\log_2 \frac{1}{2} = -1$**	**$(-1, \frac{1}{2})$**	**$(\frac{1}{2}, -1)$**
$3^2 = 9$	**$\log_3 9 = 2$**	**(2, 9)**	**(9, 2)**
$10^0 = 1$	**$\log_{10} 1 = 0$**	**(0, 1)**	**(1, 0)**

Finding e: $1 at 100% interest for 1 year

Compounded	yearly	twice	quarterly	monthly	daily	every minute
Value	$2	**$2.25**	**$2.4414**	**$2.6130**	**$2.7146**	**$2.7183**

3 Write the rule fill in each blank from what you observed

1. $\log_b y = x$ means **$b^x = y$**.

2. The graphs of $y = b^x$ and $y = \log_b x$ are reflections over **the line y = x**.

3. The domain of $y = b^x$ is **all real numbers** and its range is **y > 0**. For $y = \log_b x$ they **switch**.

4. Every graph $y = \log_b x$ passes through **(1, 0)**, because $b^0 = 1$.

5. As interest compounds more often, $(1 + \frac{1}{n})^n$ gets closer to **e** ≈ 2.718.

4 Check no tool: show your work

1. Evaluate $\log_5 125$.

   3

2. Evaluate $\log_4 \frac{1}{16}$.

   −2

3. Rewrite $10^3 = 1000$ in log form.

   $\log_{10} 1000 = 3$

4. Find the domain of $y = \log_2(x - 3)$.

   $x > 3$

5 Reflect

Why can't you take the log of a negative number?

b^x is always positive, so no exponent makes it negative; the log, its inverse, only accepts positive inputs.